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NIMCET-Free Test



By Amit Katiyar (MCA-JNU)



1. The tangent to a curve $f(x)$ makes an angle $\pi/6$ at $x=1$ and $\pi/4$ at $x=4$ with the x -axis. The value of $\int_1^4 f'(x) f''(x) dx$ is: (Integration)
- (a) $\frac{\sqrt{3}+1}{3}$ (b) $\frac{\sqrt{3}-1}{3}$ (c) 3 (d) $1/3$

NIMCET 2007

2. The value of $\int_{-2}^3 |1-x^2| dx$ is (Integration)
- (a) $7/3$ (b) $14/3$ (c) $28/3$ (d) $1/3$

NIMCET 2007

3. $\int \frac{dx}{\cos x - \sin x}$ is equal to: (Integration)

NIMCET 2007

- (a) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{3\pi}{8} \right) \right| + c$ (b) $\frac{1}{\sqrt{2}} \log \left| \cot \left(\frac{x}{2} \right) \right| + c$
 (c) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{\pi}{8} \right) \right| + c$ (d) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) \right| + c$

4. $\int \log_{10} x dx$ is **NIMCET 2010**

- (a) $(x-1) \log_e x + C$ (b) $\log_e 10 \cdot x \log_e \left(\frac{x}{e} \right) + C$
 (c) $\log_e 10 \cdot x \log_e \left(\frac{x}{e} \right) + C$ (d) $\frac{1}{x} + C$

5. The value of $\int \sqrt{x} e^{\sqrt{x}} dx$ is equal to **NIMCET 2014**

- (a) $2\sqrt{x} - e^{\sqrt{x}} - 4\sqrt{x} e^{\sqrt{x}} + C$
 (b) $(2x - 4\sqrt{x} + 4)2\sqrt{x} + C$
 (c) $(2x + 4\sqrt{x} + 4)e^{\sqrt{x}} + C$
 (d) $(1 - 4\sqrt{x})e^{\sqrt{x}} + C$

6. The value of $\int \frac{(x+1)}{x(xe^x+1)} dx$ is equal to

NIMCET 2014

- (a) $\log \frac{1+xe^x}{xe^x} + C$ (b) $\log[xe^3(1+xe^x)] + C$
 (c) $\log \left[\frac{1}{1+xe^x} \right] + C$ (d) $\log \left[\frac{xe^x}{1+xe^x} \right] + C$

7. If $\int \frac{xe^x}{\sqrt{1+xe^x}} dx = f(x)\sqrt{1+e^x} - 2 \log \frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1} + C$, then $f(x)$ is

NIMCET 2014

- (a) $2x-1$ (b) $2x-4$
 (c) $x+4$ (d) $x-4$

8. If $\int e^x (f(x) - f'(x)) dx = \phi(x)$, then the value of $\int e^x f(x) dx$ is

NIMCET 2015

- (a) $\phi(x) + e^x f(x)$ (b) $\phi(x) - e^x f(x)$
 (c) $1/2[\phi(x) + e^x f(x)]$ (d) $1/2[\phi(x) + e^x f'(x)]$

9. $\int \frac{(\log x - 1)^2}{1 + (\log x)^2} dx$ is equal to

NIMCET 2016

- (a) $\frac{xe^x}{1+x^2} + C$ (b) $\frac{x}{(\log x)^2 + 1} + C$

- (c) $\frac{\log x}{(\log x)^2 + 1} + C$ (d) $\frac{x}{x^2 + 1} + C$

10. $\int \frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}} dx$ is equal to

NIMCET 2016

- (a) $\frac{\sqrt{2x^4-2x^2+1}}{x^2} + C$ (b) $\frac{\sqrt{2x^4-2x^2+1}}{x^3} + C$

- (c) $\frac{\sqrt{2x^4-2x^2+1}}{x} + C$ (d) $\frac{\sqrt{2x^4-2x^2+1}}{2x^2} + C$

11. If $\int \cos x \cos 2x \cos 5x dx = A_1 \sin 2x + A_2 \sin 4x +$

$A_3 \sin 6x + A_4 \sin 8x + C$ then the values of A_1, A_2, A_3, A_4 are **NIMCET 2019**

- (a) $A_1 = \frac{1}{2}, A_2 = \frac{1}{4}, A_3 = \frac{1}{6}, A_4 = \frac{1}{8}$
 (b) $A_1 = \frac{1}{8}, A_2 = \frac{1}{16}, A_3 = \frac{1}{24}, A_4 = \frac{1}{32}$
 (c) $A_1 = \frac{1}{6}, A_2 = \frac{1}{12}, A_3 = \frac{1}{18}, A_4 = \frac{1}{24}$
 (d) $A_1 = \frac{1}{4}, A_2 = \frac{1}{8}, A_3 = \frac{1}{12}, A_4 = \frac{1}{16}$

12. The integral $\int \sqrt{1+2 \cot x} (\operatorname{cosec} x + \cot x) dx$ ($0 < x < \frac{\pi}{2}$), where C is constant of integration is equal to **NIMCET 2019**

- (a) $2 \log \left(\sin \frac{x}{2} \right) + C$ (b) $2 \log \left(\cos \frac{x}{2} \right) + C$
 (c) $4 \log \left(\cos \frac{x}{2} \right) + C$ (d) $4 \log \left(\sin \frac{x}{2} \right) + C$

13. If $\int \sec^2 x \operatorname{cosec}^4 x dx = -\frac{1}{3} \cot^3 x + k \tan x - 2 \cot x + C$, then the value of k is **NIMCET 2020**

- (a) 1 (b) 2 (c) 3 (d) 4

14. Evaluate $\int e^x \left(\frac{1+\sin x \cos x}{\cos^2 x} \right) dx$ **NIMCET 2020**

- (a) $e^x \cos x + C$ (b) $e^x \sec x \tan x + C$
 (c) $e^x \tan x + C$ (d) $e^x \cos 2x - 1 + C$

15. $3^{3^{3x}} 3^{3x} 3^x dx$ is equal to **NIMCET 2021**

- (a) $\frac{3^x}{(\log 3)^3} + C$ (b) $\frac{3^{3x}}{(\log 3)^3} + C$
 (c) $\frac{3^{3^{3x}}}{(\log 3)^3}$ (d) None of these

16. $\int e^x (\sin hx + \cos hx) dx = ?$ **NIMCET 2021**

- (a) $e^x \sec hx + C$ (b) $e^x \cos hx + C$
 (c) $\sin h 2x + C$ (d) $\cos h 2x + C$

17. The value of $\int \frac{(x^2-1)}{x^3\sqrt{2x^4-2x^2+1}} dx$ is **NIMCET 2022**

- (a) $2\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + C$ (b) $2\sqrt{2 + \frac{2}{x^2} + \frac{1}{x^4}} + C$
 (c) $\frac{1}{2}\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + C$ (d) None of these

18. $\int f(x) dx = g(x)$ then $\int x^5 f(x^3) dx$ **NIMCET 2023**

- (a) $\frac{1}{3} x^3 g(x^3) - 3 \int x^4 g(x^3) dx + c$
 (b) $\frac{1}{3} x^3 g(x^3) - \int x^2 g(x^3) dx + c$
 (c) $\frac{1}{3} x^3 g(x^3) - \int x^3 g(x^3) dx + c$
 (d) None of these

19. If $\int x \sin x \sec^3 x dx = \frac{1}{2} \left[f(x) \sec^2 x + g(x) \left(\frac{\tan x}{x} \right) \right] + c$ then which of the following is true **NIMCET 2023**

- (a) $f(x) - g(x) = 0$ (b) $f(x) \cdot g(x) = 0$
 (c) $f(x) + g(x) = 0$ (d) $f(x) + g(x) = 1$

ANSWER

1.	d	2.	c	3.	d	4.	*	5.	b
6.	d	7.	b	8.	c	9.	b	10.	d
11.	b	12.	a	13.	a	14.	c	15.	c
16.	b	17.	c	18.	b	19.	c		



**Analysis 2008 to 2025
Questions Asked in NIMCET 2008 to 2025**

Years	No. of Questions	Years	No. of Questions
NIMCET 2008	00	NIMCET 2016	02
NIMCET 2009	00	NIMCET 2017	00
NIMCET 2010	01	NIMCET 2018	00
NIMCET 2011	00	NIMCET 2019	02
NIMCET 2012	00	NIMCET 2020	02
NIMCET 2013	00	NIMCET 2021	02
NIMCET 2014	03	NIMCET 2022	01
NIMCET 2015	01	NIMCET 2023	02
		NIMCET 2024	00
		NIMCET 2025	00

SOLUTION

1. (d)

$$f'(1) = \tan \frac{\pi}{6}$$

$$f'(4) = \tan \frac{\pi}{4}$$

$$\int_1^4 f'(x) \cdot f''(x) dx$$

$$f'(x) = t$$

$$f''(x) dx = dt$$

$$\int_1^4 t dt \Rightarrow \left[\frac{t^2}{2} \right]_1^4$$

$$= \frac{1}{2} [(f'(x))^2]_1^4$$

$$= \frac{1}{2} [(f'(4))^2 - (f'(1))^2]$$

$$= \frac{1}{2} \left[\left(\tan \frac{\pi}{4} \right)^2 - \left(\tan \frac{\pi}{6} \right)^2 \right]$$

$$= \frac{1}{2} \left[1 - \frac{1}{3} \right]$$

$$= \frac{1}{2} \left[\frac{2}{3} \right] = \frac{1}{3}$$

2. (c) $\int_{-2}^3 |1 - x^2| dx$

$$\int_{-2}^{-1} -(1 - x^2) dx + \int_{-1}^1 (1 - x^2) dx + \int_1^3 -(1 - x^2) dx$$

$$\left[\frac{x^3}{3} - x \right]_{-2}^{-1} + \left[x - \frac{x^3}{3} \right]_{-1}^1 + \left[\frac{x^3}{3} - x \right]_1^3$$

$$= \frac{-1}{3} + 1 + \frac{8}{3} - 2 + 1 - \frac{1}{3} + 1 - \frac{1}{3} + 9 - 3 - \frac{1}{3} + 1$$

$$7 + \frac{7}{3} = \frac{28}{3}$$

3. (d) $\int \frac{dx}{\cos x - \sin x}$ is equal to

$$\frac{1}{\sqrt{2}} \int \frac{1}{\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x} dx$$

$$\frac{1}{\sqrt{2}} \int \frac{dx}{\cos \frac{\pi}{4} \cos x - \sin \frac{\pi}{4} \sin x}$$

$$\frac{1}{\sqrt{2}} \int \frac{dx}{\cos \left(\frac{\pi}{4} + x \right)}$$

$$\frac{1}{\sqrt{2}} \int \sec \left(\frac{\pi}{4} + x \right) dx$$

$$\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{\pi}{8} + \frac{\pi}{4} \right) \right| \Rightarrow \frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) \right|$$

$$= \frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} - \frac{\pi}{2} + \frac{\pi}{2} \right) \right|$$

$$\Rightarrow -\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) \right| + C$$

4. (Wrong) $\int \log_{10} x dx$

$$\text{We know, } \log_a b = \frac{\log_e b}{\log_e a}$$

$$I = \int \frac{\log_e x}{\log_e 10} dx = \frac{1}{\log_e 10} \int \log_e x dx$$

$$\text{We should know, } \int \ln x dx = [x \ln x - x] + C$$

$$= \frac{[x \ln x - x]}{\log_e 10} + C$$

$$= \log_{10} e (x \ln x - x) + C$$

$$= x \log_{10} e \log \left(\frac{x}{e} \right) + C$$

5. (b) $\int \sqrt{x} e^{\sqrt{x}} dx$

$$\text{Put } x = t^2$$

$$\Rightarrow dx = 2t dt$$

$$= \int t \cdot e^t \cdot 2t dt = \int 2t^2 \cdot e^t dt = 2 \left[\int t^2 \cdot e^t dt \right]$$

Using integration by parts,

$$= 2 \left[t^2 \int e^t dt - \int \frac{d}{dt} (t^2) \cdot \int e^t dt \right]$$

$$= 2 [t^2 \cdot e^t - \int 2t \cdot e^t dt]$$

$$= 2 [t^2 \cdot e^t - [2te^t - 2 \int e^t dt]]$$

$$= 2 [t^2 \cdot e^t - 2te^t + 2e^t] + C$$

$$= 2t^2 \cdot e^t - 4te^t + 4e^t + C$$

$$t \rightarrow \sqrt{x}$$

$$2xe^{\sqrt{x}} - 4\sqrt{x}e^{\sqrt{x}} + 4e^{\sqrt{x}} + C$$

$$= (2x - 4\sqrt{x} + 4)e^{\sqrt{x}} + C$$

6. (d) $\int \frac{x+1}{x(xe^x+1)} dx$

Substitute as $xe^x + 1 = t$

$$(xe^x + e^x) dx = dt$$

$$(x+1)e^x dx = dt$$

$$xe^x = t - 1$$

$$\int \frac{(x+1)e^x dx}{e^x x(xe^x+1)}$$

$$= \int \frac{dt}{(t-1)t}$$

$$= \int \left(\frac{1}{t-1} - \frac{1}{t} \right) dt$$

$$= \int \frac{1}{t-1} dt - \int \frac{1}{t} dt$$



$$= \log(t-1) - \log t + C$$

$$= \log xe^x - \log(xe^x + 1)$$

$$= \log\left(\frac{xe^x}{xe^x + 1}\right) + C$$

7. (b) $\int \frac{xe^x}{\sqrt{1+e^x}} dx = f(x)\sqrt{1+e^x} - 2\log\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1} + C$

$$1 + e^x = t^2 \quad e^x = t^2 - 1$$

$$e^x dx = 2t dt \quad x = \ln(t^2 - 1)$$

$$= \int \frac{x 2t dt}{t}$$

$$= 2 \int \ln(t^2 - 1) dt$$

Using integration by parts,

$$= 2 \ln(t^2 - 1)t - \int \frac{1}{t^2 - 1} 2t \cdot t dt$$

$$= 2 \left[t \cdot \ln(t^2 - 1) - 2 \int \frac{t^2 - 1 + 1}{t^2 - 1} dt \right]$$

$$= 2 \left[t \cdot \ln(t^2 - 1) - 2 \int dt - 2 \int \frac{1}{t^2 - 1} dt \right]$$

$$= 2 \left[t \cdot \ln(t^2 - 1) - 2t - 2 \ln \left| \frac{t-1}{t+1} \right| \right] + C$$

$$= 2 \left[x\sqrt{1+e^x} - 2\sqrt{1+e^x} - 2 \ln \left| \frac{\sqrt{e^x+1}-1}{\sqrt{e^x+1}+1} \right| \right] + C$$

$$= 2x\sqrt{1+e^x} - 4\sqrt{1+e^x} - 2 \ln \left| \frac{\sqrt{e^x+1}-1}{\sqrt{e^x+1}+1} \right| + C$$

$$\therefore f(x) = 2x - 4$$

8. (c) $\int e^x(f(x) - f'(x))dx = \phi(x)$

$$\int e^x f(x) dx = ?$$

$$\int e^x f(x) dx - \int e^x f'(x) dx = \phi(x)$$

Using integration by parts,

$$\int e^x f(x) dx - [e^x \int f'(x) dx - \int e^x \int f'(x) dx] = \phi(x)$$

$$\int e^x f(x) dx - e^x f(x) + \int e^x f(x) dx = \phi(x)$$

$$2 \int e^x f(x) dx = \phi(x) + e^x f(x)$$

$$\int e^x f(x) dx = \frac{1}{2} [\phi(x) + e^x f(x)]$$

9. (b) $\int \frac{(\log x - 1)^2}{1 + (\log x)^2} dx$

$$= \int \frac{(\log x - 1)^2}{(1 + (\log x)^2)^2} dx = \int \frac{(\log x)^2 + 1 - 2 \log x}{(1 + (\log x)^2)^2} dx$$

$$= \int \frac{1}{(1 + (\log x)^2)} - \int \frac{2 \log x}{(1 + (\log x)^2)^2}$$

Using integration by parts in first integral,

$$\int (1 + (\log x)^2)^{-1} - \int \frac{2 \log x}{(1 + (\log x)^2)^2}$$

$$= x \cdot (1 + (\log x)^2)^{-1} - \int (-1)(1 + (\log x)^2)^{-2} \times \frac{2 \log x}{x} \cdot x$$

$$- \int \frac{2 \log x}{(1 + (\log x)^2)^2}$$

$$= \frac{x}{(1 + (\log x)^2)} + \int \frac{2 \log x}{(1 + (\log x)^2)^2} - \int \frac{2 \log x}{(1 + (\log x)^2)^2} + C$$

$$= \frac{x}{1 + (\log x)^2} + C$$

10. (d) $\int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}} = \int \frac{x^2 - 1}{x^3 \sqrt{x^4(2 - \frac{2}{x^2} + \frac{1}{x^4})}}$

$$= \int \frac{x^2 - 1}{x^5 \sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}}}$$

Now, substitute

$$2 - \frac{2}{x^2} + \frac{1}{x^4} = t$$

$$\Rightarrow 2 - 2x^{-2} + x^{-4} = t$$

$$\Rightarrow (0 - 2(-2)x^{-3} - 4x^{-5})dx = dt$$

$$\Rightarrow (4x^{-3} - 4x^{-5})dx = dt \Rightarrow (x^{-3} - x^{-5})dx = \frac{dt}{4}$$

Then,

$$= \int \frac{x^{-3} - x^{-5}}{\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}}} = \int \frac{dt/4}{\sqrt{t}} = \frac{1}{4} \int \frac{dt}{\sqrt{t}} = \frac{1}{4} \times 2\sqrt{t}$$

$$= \frac{1}{2} \sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + C = \frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + C$$

11. (b) If $\int \cos x \cos 2x \cos 5x dx = A_1 \sin 2x + A_2 \sin 4x$

$$+ A_3 \sin 6x + A_4 \sin 8x + C$$

$$\int \cos x \times \frac{1}{2} (2 \cos 2x \cos 5x) \cdot dx$$

Using $2 \cos A \cdot \cos B = \cos(A+B) + \cos(A-B)$,

$$\frac{1}{2} \int \cos x [\cos 7x + \cos 3x] dx$$

$$= \frac{1}{2} \int \cos x \cos 7x dx + \frac{1}{2} \int \cos x \cos 3x dx$$

$$= \frac{1}{4} \int 2 \cos x \cos 7x dx + \frac{1}{4} \int 2 \cos x \cos 3x dx$$

$$= \frac{1}{4} \int [\cos 8x + \cos 6x + \cos 4x + \cos 2x] dx$$

$$= \frac{1}{4} \left[\int \cos 8x dx + \int \cos 6x dx + \int \cos 4x dx + \int \cos 2x dx \right]$$

$$\text{Using } \int \cos ax dx = \frac{\sin ax}{a} + C$$

$$= \frac{1}{4} \left[\frac{\sin 8x}{8} + \frac{\sin 6x}{6} + \frac{\sin 4x}{4} + \frac{\sin 2x}{2} \right] + C$$

$$= \frac{\sin 8x}{32} + \frac{\sin 6x}{24} + \frac{\sin 4x}{16} + \frac{\sin 2x}{8} + C$$

$$= \frac{\sin 2x}{8} + \frac{\sin 4x}{16} + \frac{\sin 6x}{24} + \frac{\sin 8x}{32} + C$$



$A_1 = \frac{1}{8}$	$A_2 = \frac{1}{16}$	$A_3 = \frac{1}{24}$	$A_4 = \frac{1}{32}$
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12. (a) $\int \sqrt{1 + 2\cot x(\operatorname{cosec} x + \cot x)} dx, (0 < x < \frac{\pi}{2})$
 $\therefore \int (\sqrt{\operatorname{cosec}^2 x - \cot^2 x + 2\cot x \operatorname{cosec} x + 2\cot^2 x}) dx$
 $= \int (\sqrt{(\operatorname{cosec} x + \cot x)^2}) dx = \int (\operatorname{cosec} x + \cot x) dx$
 $= \int \left(\frac{1+\cos x}{\sin x} \right) dx = \int \frac{2\cos^2 \frac{x}{2}}{2\sin \frac{x}{2} \cos \frac{x}{2}} dx$
 $= \int \cot \frac{x}{2} dx = \frac{\log(\sin \frac{x}{2})}{\frac{1}{2}} + C = 2\log\left(\sin \frac{x}{2}\right) + C$

13. (a)
 $\int \sec^2 x \operatorname{cosec}^4 x dx = -\frac{1}{3} \cot^3 x + k \tan x - 2\cot x + C$
 $\int (1 + \tan^2 x) \operatorname{cosec}^2 x \cdot \operatorname{cosec}^2 x dx$
 $= \int \left(1 + \frac{1}{\cot^2 x}\right) (1 + \cot^2 x) \cdot \operatorname{cosec}^2 x dx$
 let $\cot x = t$
 $\Rightarrow -\operatorname{cosec}^2 x dx = dt \Rightarrow \operatorname{cosec}^2 x dx = -dt$
 $= \int \left(1 + \frac{1}{t^2}\right) (1 + t^2) (-dt) = -\int \left(1 + \frac{1}{t^2}\right) (1 + t^2) dt$
 $= -\left[\int \left(1 + \frac{1}{t^2} + t^2 + 1\right) dt\right] = -\left[\int 2 + \int t^{-2} dt + \int t^2 dt\right]$
 $= -\left[2t + \frac{t^{-2+1}}{-2+1} + \frac{t^3}{3}\right] + C = -\left[2\cot x - \frac{1}{\cot x} + \frac{\cot^3 x}{3}\right] + C$
 $= \frac{-\cot^3 x}{3} + \tan x - 2\cot x + C \therefore k = 1$

14. (c) $\int e^x \left(\frac{1+\sin x \cos x}{\cos^2 x} \right) dx = \int e^x (\sec^2 x + \tan x) dx$

Using formula,

$$\int e^x (f(x) + f'(x)) dx = e^x f(x) + C$$

Here, $f(x) = \tan x, f'(x) = \sec^2 x$

Hence, its answer is $e^x \tan x + C$

15. (c) $\int 3^{3^{3^x}} 3^{3^x} 3^x dx$

Put $3^{3^{3^x}} = t \Rightarrow 3^{3^{3^x}} \log 3 \cdot 3^{3^x} \log 3 \cdot 3^x \log 3 \cdot dx = dt$

$$3^{3^{3^x}} \cdot 3^{3^x} \cdot 3^x dx = \frac{dt}{(\log 3)^3}$$

$$= \int \frac{dt}{(\log 3)^3} = \frac{t}{(\log 3)^3} + C = \frac{3^{3^{3^x}}}{(\log 3)^3} + C$$

16. (b) $\int e^x (\sin hx + \cos hx) dx$

$$\sin hx = \frac{e^x - e^{-x}}{2}, \cos hx = \frac{e^x + e^{-x}}{2}, \frac{d}{dx} \cos hx = \sin hx$$

Using formula,

$$\int e^x (f(x) + f'(x)) dx = e^x f(x) + C$$

$$\int e^x (\cos hx + \sin hx) dx = e^x \cos hx + C$$

14. (c) $\int \frac{(x^2-1)dx}{x^3 \sqrt{2x^4-2x^2+1}} = \int \frac{(x^2-1)dx}{x^3 \sqrt{x^4(2-\frac{2}{x^2}+\frac{1}{x^4})}}$

$$= \int \frac{(x^2-1)dx}{x^5 \sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}}$$

Now substitute, $2 - \frac{2}{x^2} + \frac{1}{x^4} = t, 2 - 2x^{-2} + x^{-4} = t$

$$(0 - 2(-2)x^{-3} - 4x^{-5}) dx = dt \Rightarrow (4x^{-3} - 4x^{-5}) dx = dt$$

$$(x^{-3} - x^{-5}) dx = dt/4$$

$$\therefore \int \frac{(x^{-3}-x^{-5})dx}{\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}}$$

$$= \int \frac{dt/4}{\sqrt{t}} = \frac{1}{4} \int \frac{dt}{\sqrt{t}} = \frac{1}{4} \times 2\sqrt{t} = \frac{1}{2} \sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + C$$

17. (c)

18. (b) $I = \int x^5 f(x^3) dx$

$$= \int x^2 x^3 f(x^3) dx$$

Let $x^3 = t$

$$\Rightarrow 3x^2 dx = dt$$

$$I = \frac{1}{3} \int t f(t) dt$$

Using integration by parts.

$$I = \frac{1}{3} \left[t \int f(t) dt - \int \left(\frac{dt}{dt} \int f(t) dt \right) dt \right]$$

$$= \frac{1}{3} \left[t g(t) - \int g(t) dt \right]$$

$$= \frac{1}{3} t g(t) - \frac{1}{3} \int g(t) dt$$

As $x^3 = t$ and $dt = 3x^2 dx$

$$= \frac{1}{3} x^3 g(x^3) - \frac{3}{3} \int x^2 g(x^3) dx$$

$$= \frac{1}{3} x^3 g(x^3) - \int x^2 g(x^3) dx + c$$

19. (c) $\int x \sin x \sec^3 x dx = \int x \tan x \sec^2 x dx$

Using integration by part

$$= [x \int \sec^2 x \tan x dx] - \int \left(\frac{dx}{dx} \int \sec^2 x \tan x dx \right) dx$$

$$= \left[x \frac{\tan^2 x}{2} \right] - \int \frac{\tan^2 x}{2} dx$$

$$= \frac{1}{2} [x \tan^2 x - \int (\sec^2 x - 1) dx]$$

$$= \frac{1}{2} [x \tan^2 x - \tan x + x] + c$$



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$$= \frac{1}{2} [x(1 + \tan^2 x) - \tan x] + c$$

$$= \frac{1}{2} [x \sec^2 x - \tan x] + c$$

$$= \frac{1}{2} \left[f(x) \sec^2 x + g(x) \left(\frac{\tan x}{x} \right) \right] + c$$

$$\text{Hence, } f(x) = x; g(x) = -x$$

$$\text{Hence, } f(x) + g(x) = 0$$



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